Meen Patel

Nuremberg, Bavaria, Germany

About

M.Sc. candidate in Autonomy Technologies at Friedrich Alexander Universität Erlangen Nürnberg with hands-on experience in embedded firmware development, ESP32-based AI systems, and drone vision modules. Experienced in real-time systems, microcontroller programming, and IoT integration.

Skills

Languages: C, Embedded C, Python, Assembly (MASM), MATLAB
Embedded Platforms: ESP32, STM32, 8051, Arduino, ARM Cortex-M, DMA, Timers, PWM, ADC, DAC
Protocols: UART, SPI, I2C, MQTT
RTOS & Firmware: FreeRTOS, Zephyr, State Machines, Interrupt Handling, Task Scheduling, Software Debugging, Firmware-Hardware Integration
Tools: Git/GitHub, Keil, Proteus, ModelSim, Espressif IDE, MATLAB
Hardware: Oscilloscope, Multimeter, Sensor Integration, Sensor Calibration, Prototype Assembly
Computer Vision: OpenCV, Image Processing
Domains: Embedded Systems, IoT, Industrial Automation, Digital Electronics, Robotics, Software Development
Robotics & Autonomy: Sensor Fusion, ROS2, Autonomous Navigation, Path Planning, Computer Vision

Contact

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Education

Friedrich-Alexander-Universität Erlangen-Nürnberg — **2026 – Present**
M.Sc. in Autonomy Technologies — *Erlangen, Germany*

Nirma University — **2021 – 2025**
B.Tech in ECE — *Ahmedabad, Gujarat*

Vishwabharati English Medium School — **2020 – 2021**
12th — *Ahmedabad, Gujarat*

Experience

NXON AI Pvt. Ltd. | *Microcontrollers, espressif* — **Dec. 2024 – Jan. 2026**
Junior Embedded Engineer — *April 2025 – Jan. 2026*
Embedded Software Developer Intern — *Dec. 2024 – April 2025*

- Designed and developed a secure lock system using nRF52840 with NFC and BLE connectivity with proper authentication.
- Assigned key responsibilities included customizing and enhancing open-source firmware based on specific client requirements and NRF boards, encompassing user interaction features and system performance optimization.
- Developed a Gemini-powered project on ESP32 with speaker and microphone integration.
- Contributed to the development of a drone vision module, assisting in camera integration and real-time processing under senior engineering supervision.

eInfochips - An ARROW Company | *Research* — **Jun. 2024 – Jul. 2024**
R & D Intern

- Conducted research on AI applications in embedded electronics and wireless IoT systems, analyzing product architecture and connectivity solutions.

Midas Metcons Pvt. Ltd. |*Embedded C, 8051* — **Jun. 2023 – Dec. 2023**
Embedded Intern

- Developed Embedded C firmware for microcontroller-based automated storage systems and performed simulation-based validation.

Projects

Secure Lock System using nRF52840
- Implemented low-latency speech processing with cloud API integration and optimized firmware for stable real-time interaction via NFC and BLE.

ESP32 AI Voice Assistant
- Implemented low-latency speech processing with cloud API integration and optimized firmware for stable real-time interaction.

Vertical Lift Module (Warehouse Automation)
- Designed RTOS-based task scheduling system for efficient actuator control and IoT-based remote diagnostics.

Vision Control for Drone System
- Calibrate camera to correct distortion and detect marker, estimate pose, and calculate movement.

MIPS Assembler in Python
- Implemented a script that translates assembly instructions into machine code with error handling to demonstrate assembler functionality.

Image Modulation Using Sound (MATLAB)
- Dynamically changed image based on sound frequency and amplitude.

**For more projects, visit my portfolio: meenpatel.github.io/portfolio **

Achievements

AMD Pervasive AI Developer Contest — **Feb 2024 – Aug 2024**
Neuro Glove: Smart Rehabilitation Assistant — *Hackster.io*

Smart India Hackathon — **Sep 2023**
Regional Level Finalist — *E-Cell, Nirma University*